

10

Linear Fractional Transformation

This chapter introduces a new matrix function: linear fractional transformation (LFT). We show that many interesting control problems can be formulated in an LFT framework and thus can be treated using the same technique.

10.1 Linear Fractional Transformations

This section introduces the matrix linear fractional transformations. It is well known from the one complex variable function theory that a mapping $F : \mathbb{C} \mapsto \mathbb{C}$ of the form

$$F(s) = \frac{a + bs}{c + ds}$$

with a, b, c and $d \in \mathbb{C}$ is called a *linear fractional transformation*. In particular, if $c \neq 0$ then $F(s)$ can also be written as

$$F(s) = \alpha + \beta s(1 - \gamma s)^{-1}$$

for some α, β and $\gamma \in \mathbb{C}$. The linear fractional transformation described above for scalars can be generalized to the matrix case.

Definition 10.1 Let M be a complex matrix partitioned as

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \in \mathbb{C}^{(p_1 + p_2) \times (q_1 + q_2)},$$

and let $\Delta_\ell \in \mathbb{C}^{q_2 \times p_2}$ and $\Delta_u \in \mathbb{C}^{q_1 \times p_1}$ be two other complex matrices. Then we can formally define a *lower LFT* with respect to Δ_ℓ as the map

$$\mathcal{F}_\ell(M, \bullet) : \mathbb{C}^{q_2 \times p_2} \mapsto \mathbb{C}^{p_1 \times q_1}$$

with

$$\mathcal{F}_\ell(M, \Delta_\ell) \triangleq M_{11} + M_{12}\Delta_\ell(I - M_{22}\Delta_\ell)^{-1}M_{21}$$

provided that the inverse $(I - M_{22}\Delta_\ell)^{-1}$ exists. We can also define an *upper LFT* with respect to Δ_u as

$$\mathcal{F}_u(M, \bullet) : \mathbb{C}^{q_1 \times p_1} \mapsto \mathbb{C}^{p_2 \times q_2}$$

with

$$\mathcal{F}_u(M, \Delta_u) = M_{22} + M_{21}\Delta_u(I - M_{11}\Delta_u)^{-1}M_{12}$$

provided that the inverse $(I - M_{11}\Delta_u)^{-1}$ exists.

The matrix M in the above LFTs is called the *coefficient matrix*. The motivation for the terminologies of *lower* and *upper* LFTs should be clear from the following diagram representations of $\mathcal{F}_\ell(M, \Delta_\ell)$ and $\mathcal{F}_u(M, \Delta_u)$:



The diagram on the left represents the following set of equations:

$$\begin{bmatrix} z_1 \\ y_1 \end{bmatrix} = M \begin{bmatrix} w_1 \\ u_1 \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} w_1 \\ u_1 \end{bmatrix},$$

$$u_1 = \Delta_\ell y_1$$

while the diagram on the right represents

$$\begin{bmatrix} y_2 \\ z_2 \end{bmatrix} = M \begin{bmatrix} u_2 \\ w_2 \end{bmatrix} = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix} \begin{bmatrix} u_2 \\ w_2 \end{bmatrix},$$

$$u = \Delta_u y_2.$$

It is easy to verify that the mapping defined on the left diagram is equal to $\mathcal{F}_\ell(M, \Delta_\ell)$ and the mapping defined on the right diagram is equal to $\mathcal{F}_u(M, \Delta_u)$. So from the above diagrams, $\mathcal{F}_\ell(M, \Delta_\ell)$ is a transformation obtained from closing the *lower* loop on the left diagram; similarly $\mathcal{F}_u(M, \Delta_u)$ is a transformation obtained from closing the *upper* loop on the right diagram. In most cases, we will use the general term *LFT* in referring to both upper and lower LFTs and assume that the contents will distinguish the situations. The reason for this is that one can use either of these notations to express

a given object. Indeed, it is clear that $\mathcal{F}_u(N, \Delta) = \mathcal{F}_\ell(M, \Delta)$ with $N = \begin{bmatrix} M_{22} & M_{21} \\ M_{12} & M_{11} \end{bmatrix}$.

It is usually not crucial which expression is used; however, it is often the case that one expression is more convenient than the other for a given problem. It should also be clear to the reader that in writing $\mathcal{F}_\ell(M, \Delta)$ (or $\mathcal{F}_u(M, \Delta)$) it is implied that Δ has compatible dimensions.

A useful interpretation of an LFT, e.g., $\mathcal{F}_\ell(M, \Delta)$, is that $\mathcal{F}_\ell(M, \Delta)$ has a nominal mapping, M_{11} , and is perturbed by Δ , while M_{12} , M_{21} , and M_{22} reflect a prior knowledge as to how the perturbation affects the nominal map, M_{11} . A similar interpretation can be applied to $\mathcal{F}_u(M, \Delta)$. This is why LFT is particularly useful in the study of perturbations, which is the focus of the next chapter.

The physical meaning of an LFT in control science is obvious if we take M as a proper transfer matrix. In that case, the LFTs defined above are simply the closed-loop transfer matrices from $w_1 \mapsto z_1$ and $w_2 \mapsto z_2$, respectively, i.e.,

$$T_{zw1} = \mathcal{F}_\ell(M, \Delta_\ell), \quad T_{zw2} = \mathcal{F}_u(M, \Delta_u)$$

where M may be the controlled plant and Δ may be either the system model uncertainties or the controllers.

Definition 10.2 An LFT, $\mathcal{F}_\ell(M, \Delta)$, is said to be *well defined (or well-posed)* if $(I - M_{22}\Delta)$ is invertible.

Note that this definition is consistent with the well-posedness definition of the feedback system, which requires the corresponding transfer matrix be invertible in $\mathcal{R}_p(s)$. It is clear that the study of an LFT that is not well-defined is meaningless, hence throughout the book, whenever an LFT is invoked, it will be assumed implicitly to be well defined. It is also clear from the definition that, for any M , $\mathcal{F}_\ell(M, 0)$ is well defined; hence any function that is not well defined at the origin cannot be expressed as an LFT in terms of its variables. For example, $f(\delta) = 1/\delta$ is not an LFT of δ .

In some literature, LFT is used to refer to the following matrix functions:

$$(A + BQ)(C + DQ)^{-1} \quad \text{or} \quad (C + QD)^{-1}(A + QB)$$

where C is usually assumed to be invertible due to practical consideration. The following is true.

Lemma 10.1 Suppose C is invertible. Then

$$\begin{aligned} (A + BQ)(C + DQ)^{-1} &= \mathcal{F}_\ell(M, Q) \\ (C + QD)^{-1}(A + QB) &= \mathcal{F}_\ell(N, Q) \end{aligned}$$

with

$$M = \begin{bmatrix} AC^{-1} & B - AC^{-1}D \\ C^{-1} & -C^{-1}D \end{bmatrix}, \quad N = \begin{bmatrix} C^{-1}A & C^{-1} \\ B - DC^{-1}A & -DC^{-1} \end{bmatrix}.$$

The converse also holds if M satisfies certain conditions.

Lemma 10.2 *Let $\mathcal{F}_\ell(M, Q)$ be a given LFT with $M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$, then*

(a) *if M_{21} is invertible,*

$$\mathcal{F}_\ell(M, Q) = (A + BQ)(C + DQ)^{-1}$$

with

$$A = M_{11}M_{21}^{-1}, \quad B = M_{12} - M_{11}M_{21}^{-1}M_{22}, \quad C = M_{21}^{-1}, \quad D = -M_{21}^{-1}M_{22}.$$

(b) *if M_{12} is invertible,*

$$\mathcal{F}_\ell(M, Q) = (C + QD)^{-1}(A + QB)$$

with

$$A = M_{12}^{-1}M_{11}, \quad B = M_{21} - M_{22}M_{12}^{-1}M_{11}, \quad C = M_{12}^{-1}, \quad D = -M_{22}M_{12}^{-1}.$$

However, for an arbitrary LFT $\mathcal{F}_\ell(M, Q)$, neither M_{21} nor M_{12} is necessarily square and invertible; therefore, the alternative fractional formula is more restrictive.

It should be pointed out that some seemingly similar functions do not have simple LFT representations. For example,

$$(A + QB)(I + QD)^{-1}$$

cannot always be written in the form of $\mathcal{F}_\ell(M, Q)$ for some M ; however, it can be written as

$$(A + QB)(I + QD)^{-1} = \mathcal{F}_\ell(N, \Delta)$$

with

$$N = \left[\begin{array}{c|c|c} A & I & A \\ \hline -B & 0 & -B \\ \hline D & 0 & D \end{array} \right], \quad \Delta = \begin{bmatrix} Q & \\ & Q \end{bmatrix}.$$

Note that the dimension of Δ is twice as many as Q .

The following lemma summarizes some of the algebraic properties of LFTs.

Lemma 10.3 *Let M, Q , and G be suitably partitioned matrices:*

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}, \quad Q = \begin{bmatrix} Q_{11} & Q_{12} \\ Q_{21} & Q_{22} \end{bmatrix}, \quad G = \begin{bmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{bmatrix}.$$

(i) $\mathcal{F}_u(M, \Delta) = \mathcal{F}_l(N, \Delta)$ with

$$N = \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} M \begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix} = \begin{bmatrix} M_{22} & M_{21} \\ M_{12} & M_{11} \end{bmatrix}$$

where the dimensions of identity matrices are compatible with the partitions of M and N .

(ii) Suppose $\mathcal{F}_u(M, \Delta)$ is square and well-defined and M_{22} is nonsingular. Then the inverse of $\mathcal{F}_u(M, \Delta)$ exists and is also an LFT with respect to Δ :

$$(\mathcal{F}_u(M, \Delta))^{-1} = \mathcal{F}_u(N, \Delta)$$

with N given by

$$N = \begin{bmatrix} M_{11} - M_{12}M_{22}^{-1}M_{21} & -M_{12}M_{22}^{-1} \\ M_{22}^{-1}M_{21} & M_{22}^{-1} \end{bmatrix}.$$

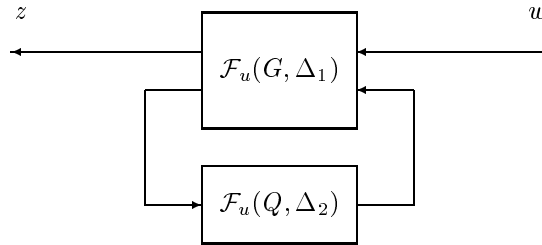
(iii) $\mathcal{F}_u(M, \Delta_1) + \mathcal{F}_u(Q, \Delta_2) = \mathcal{F}_u(N, \Delta)$ with

$$N = \begin{bmatrix} M_{11} & 0 & \vdots & M_{12} \\ 0 & Q_{11} & \vdots & Q_{12} \\ \hline M_{21} & Q_{21} & \vdots & M_{22} + Q_{22} \end{bmatrix}, \quad \Delta = \begin{bmatrix} \Delta_1 & 0 \\ 0 & \Delta_2 \end{bmatrix}.$$

(iv) $\mathcal{F}_u(M, \Delta_1)\mathcal{F}_u(Q, \Delta_2) = \mathcal{F}_u(N, \Delta)$ with

$$N = \begin{bmatrix} M_{11} & M_{12}Q_{21} & \vdots & M_{12}Q_{22} \\ 0 & M_{11} & \vdots & Q_{12} \\ \hline M_{21} & M_{22}Q_{21} & \vdots & M_{22}Q_{22} \end{bmatrix}, \quad \Delta = \begin{bmatrix} \Delta_1 & 0 \\ 0 & \Delta_2 \end{bmatrix}.$$

(v) Consider the following interconnection structure where the dimensions of Δ_1 are compatible with A :



Then the mapping from w to z is given by

$$\mathcal{F}_l(\mathcal{F}_u(G, \Delta_1), \mathcal{F}_u(Q, \Delta_2)) = \mathcal{F}_u(\mathcal{F}_l(G, \mathcal{F}_u(Q, \Delta_2)), \Delta_1) = \mathcal{F}_u(N, \Delta)$$

$$N = \left[\begin{array}{cc|c} A + B_2 Q_{22} L_1 C_2 & B_2 L_2 Q_{21} & B_1 + B_2 Q_{22} L_1 D_{21} \\ Q_{12} L_1 C_2 & Q_{11} + Q_{12} L_1 D_{22} Q_{21} & Q_{12} L_1 D_{21} \\ \hline C_1 + D_{12} L_2 Q_{22} C_2 & D_{12} L_2 Q_{21} & D_{11} + D_{12} Q_{22} L_1 D_{21} \end{array} \right]$$

$$\text{where } L_1 := (I - D_{22} Q_{22})^{-1}, L_2 := (I - Q_{22} D_{22})^{-1}, \text{ and } \Delta = \begin{bmatrix} \Delta_1 & 0 \\ 0 & \Delta_2 \end{bmatrix}.$$

Proof. These properties can be straightforwardly verified by the definition of *LFT*, so the proofs are omitted. $\square$

Property (v) shows that if the open-loop system parameters are LFTs of some variables, then the closed-loop system parameters are also LFTs of the same variables. This property is particularly useful in perturbation analysis and in building the interconnection structure. Similar results can be stated for lower LFT. It is usually convenient to interpret an LFT $\mathcal{F}_u(M, \Delta)$ as a state space realization of a generalized system with frequency structure Δ . In fact, all the above properties can be reduced to the standard transfer matrix operations if $\Delta = \frac{1}{s}I$.

The following proposition is concerned with the algebraic properties of *LFTs* in the general control setup.

Lemma 10.4 Let $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$ and let K be rational transfer function matrices and let $G = \mathcal{F}_\ell(P, K)$. Then

- (a) G is proper if P and K are proper with $\det(I - P_{22}K)(\infty) \neq 0$.
- (b) $\mathcal{F}_\ell(P, K_1) = \mathcal{F}_\ell(P, K_2)$ implies that $K_1 = K_2$ if P_{12} and P_{21} have normal full column and row rank in $\mathcal{R}_p(s)$, respectively.
- (c) If P and G are proper, $\det P(\infty) \neq 0$, $\det \left(P - \begin{bmatrix} G & 0 \\ 0 & 0 \end{bmatrix} \right)(\infty) \neq 0$ and P_{12} and P_{21} are square and invertible for almost all s , then K is proper and

$$K = \mathcal{F}_u(P^{-1}, G).$$

Proof.

- (a) is immediate from the definition of $\mathcal{F}_\ell(P, K)$ (or well-posedness condition).
- (b) follows from the identity

$$\mathcal{F}_\ell(P, K_1) - \mathcal{F}_\ell(P, K_2) = P_{12}(I - K_2 P_{22})^{-1}(K_1 - K_2)(I - P_{22} K_1)^{-1} P_{21}.$$

- (c) it is sufficient to show that the formula for K is well-posed and K thus obtained is proper. Let $Q = P^{-1}$, which will be proper since $\det P(\infty) \neq 0$, and define

$$K = \mathcal{F}_u(Q, G) = Q_{22} + Q_{21}G(I - Q_{11}G)^{-1}Q_{12}.$$

This expression is well-posed and proper since at $s = \infty$

$$\begin{aligned} \det(I - Q_{11}G) &= \det\left(I - \begin{bmatrix} I & 0 \end{bmatrix} P^{-1} \begin{bmatrix} I \\ 0 \end{bmatrix} G\right) \\ &= \det\left[P^{-1} \left(P - \begin{bmatrix} G & 0 \\ 0 & 0 \end{bmatrix}\right)\right] \neq 0. \end{aligned}$$

We also need to ensure that $\mathcal{F}_\ell(P, K)$ is well-posed:

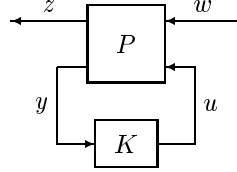
$$\begin{aligned} I - P_{22}K &= (I - P_{22}Q_{22}) - P_{22}Q_{21}G(I - Q_{11}G)^{-1}Q_{12} \\ &= P_{21}Q_{12} + P_{21}Q_{11}G(I - Q_{11}G)^{-1}Q_{12} \\ &= P_{21}(I - Q_{11}G)^{-1}Q_{12}. \end{aligned}$$

The above form is obtained by using the fact that $PQ = I$. Then $\det(I - P_{22}K) \neq 0$ since P_{21}^{-1} exists and $Q_{12}^{-1} = P_{21} - P_{22}P_{12}^{-1}P_{11}$. Hence the LFTs are both well-posed and we immediately obtain that $\mathcal{F}_\ell(P, K) = G$ as required upon substituting for K and $(I - P_{22}K)$, as shown above.

□

Remark 10.1 This lemma shows that under certain conditions, an LFT of transfer matrices is a bijective map between two sets of proper and real rational matrices. When given proper transfer matrices P and G with compatible dimensions which satisfy conditions in (c), there exists a unique proper K such that $G = \mathcal{F}_\ell(P, K)$. On the other hand, the conditions of part (c) show that the feedback systems are well-posed. ♡

Remark 10.2 A simple interpretation of the result (c) is given by considering the signals in the feedback systems,



assuming this structure is well-posed. And we have

$$\begin{aligned} \begin{bmatrix} z \\ y \end{bmatrix} &= P \begin{bmatrix} w \\ u \end{bmatrix}, \quad u = Ky \\ \Rightarrow z &= \mathcal{F}_\ell(P, K)w = Gw; \end{aligned}$$

hence

$$\begin{aligned} \begin{bmatrix} w \\ u \end{bmatrix} &= P^{-1} \begin{bmatrix} z \\ y \end{bmatrix}, \quad z = Gw \\ \Rightarrow u &= \mathcal{F}_u(P^{-1}, G)y, \quad \text{or } K = \mathcal{F}_u(P^{-1}, G). \end{aligned}$$

♡

10.2 Examples of LFTs

LFT is a very convenient tool to formulate many mathematical objects. In this section and the sections to follow, some commonly encountered control or mathematical objects are given new perspectives, i.e., they will be examined from the LFT point of view.

Polynomials

A very commonly encountered object in control and mathematics is a polynomial function. For example,

$$p(\delta) = a_0 + a_1\delta + \cdots + a_n\delta^n$$

with indeterminate δ . It is easy to verify that $p(\delta)$ can be written in the following LFT form:

$$p(\delta) = \mathcal{F}_\ell(M, \delta I_n)$$

with

$$M = \left[\begin{array}{c|cccc} a_0 & a_1 & \cdots & a_{n-1} & a_n \\ \hline 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ 0 & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{array} \right].$$

Hence every polynomial is a linear fraction of their indeterminates. More generally, any multivariate (matrix) polynomials are also LFTs in their indeterminates; for example,

$$p(\delta_1, \delta_2) = a_1\delta_1^2 + a_2\delta_2^2 + a_3\delta_1\delta_2 + a_4\delta_1 + a_5\delta_2 + a_6.$$

Then

$$p(\delta_1, \delta_2) = \mathcal{F}_\ell(N, \Delta)$$

with

$$N = \left[\begin{array}{c|cccc} a_6 & 1 & 0 & 1 & 0 \\ \hline a_4 & 0 & a_1 & 0 & a_3 \\ 1 & 0 & 0 & 0 & 0 \\ a_5 & 0 & 0 & 0 & a_2 \\ 1 & 0 & 0 & 0 & 0 \end{array} \right], \quad \Delta = \begin{bmatrix} \delta_1 I_2 & \\ & \delta_2 I_2 \end{bmatrix}.$$

It should be noted that these representations or realizations of polynomials are neither unique nor necessarily minimal. Here a minimal realization refers to a realization with the smallest possible dimension of Δ . As commonly known, in multidimensional systems and filter theory, it is usually very hard, if not impossible, to find a minimal realization for even a two variable polynomial. In fact, the minimal dimension of Δ depends also on the field (real, complex, etc.) of the realization. More detailed discussion of this issue is beyond the scope of this book, the interested readers should consult the references in 2-d or n-d systems or filter theory.

Rational Functions

As another example of LFT representation, we consider a rational matrix function (not necessarily proper), $F(\delta_1, \delta_2, \dots, \delta_m)$, with a finite value at the origin: $F(0, 0, \dots, 0)$ is finite. Then $F(\delta_1, \delta_2, \dots, \delta_m)$ can be written as an LFT in $(\delta_1, \delta_2, \dots, \delta_m)$ (some δ_i may be repeated). To see that, write

$$F(\delta_1, \delta_2, \dots, \delta_m) = \frac{N(\delta_1, \delta_2, \dots, \delta_m)}{d(\delta_1, \delta_2, \dots, \delta_m)} = N(\delta_1, \delta_2, \dots, \delta_m) (d(\delta_1, \delta_2, \dots, \delta_m)I)^{-1}$$

where $N(\delta_1, \delta_2, \dots, \delta_m)$ is a multivariate matrix polynomial and $d(\delta_1, \delta_2, \dots, \delta_m)$ is a scalar multivariate polynomial with $d(0, 0, \dots, 0) \neq 0$. Both N and dI can be represented as LFTs, and, furthermore, since $d(0, 0, \dots, 0) \neq 0$, the inverse of dI is also an LFT as shown in Lemma 10.3. Now the conclusion follows by the fact that the product of LFTs is also an LFT. (Of course, the above LFT representation problem is exactly the problem of state space realization for a multidimensional discrete transfer matrix.) However, this is usually not a nice way to get an LFT representation for a rational matrix since this approach usually results in a much higher dimensioned Δ than required. For example,

$$f(\delta) = \frac{\alpha + \beta\delta}{1 + \gamma\delta} = \mathcal{F}_\ell(M, \delta)$$

with

$$M = \left[\begin{array}{c|c} \alpha & \beta - \alpha\gamma \\ \hline 1 & -\gamma \end{array} \right].$$

By using the above approach, we would end up with

$$f(\delta) = \mathcal{F}_\ell(N, \delta I_2)$$

and

$$N = \left[\begin{array}{c|cc} \alpha & \beta & -\alpha\gamma \\ \hline 1 & 0 & -\gamma \\ 1 & 0 & -\gamma \end{array} \right].$$

Although the latter can be reduced to the former, it is not easy to see how to carry out such reduction for a complicated problem, even if it is possible.

State Space Realizations

We can use the LFT formulae to establish the relationship between transfer matrices and their state space realizations. A system with a state space realization as

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned}$$

has a transfer matrix of

$$G(s) = D + C(sI - A)^{-1}B = \mathcal{F}_u\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}, \frac{1}{s}I\right).$$

Now take $\Delta = \frac{1}{s}I$, the transfer matrix can be written as

$$G(s) = \mathcal{F}_u\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}, \Delta\right).$$

More generally, consider a discrete time 2-D (or MD) system realized by the first-order state space equation

$$\begin{aligned} x_1(k_1 + 1, k_2) &= A_{11}x_1(k_1, k_2) + A_{12}x_2(k_1, k_2) + B_1u(k_1, k_2) \\ x_2(k_1, k_2 + 1) &= A_{21}x_1(k_1, k_2) + A_{22}x_2(k_1, k_2) + B_2u(k_1, k_2) \\ y(k_1, k_2) &= C_1x_1(k_1, k_2) + C_2x_2(k_1, k_2) + Du(k_1, k_2). \end{aligned}$$

In the same way, take

$$\Delta = \begin{bmatrix} z_1^{-1}I & 0 \\ 0 & z_2^{-1}I \end{bmatrix} =: \begin{bmatrix} \delta_1 I & 0 \\ 0 & \delta_2 I \end{bmatrix}$$

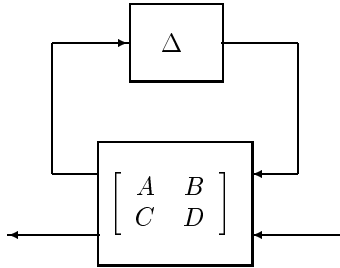
where z_i denotes the forward shift operator, and let

$$A \triangleq \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B \triangleq \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}, \quad C \triangleq [C_1 \quad C_2]$$

then its transfer matrix is

$$\begin{aligned} G(z_1, z_2) &= D + C\left(\begin{bmatrix} z_1 I & 0 \\ 0 & z_2 I \end{bmatrix} - A\right)^{-1}B = D + C\Delta(I - \Delta A)^{-1}B \\ &=: \mathcal{F}_u\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}, \Delta\right). \end{aligned}$$

Both formulations can correspond to the following diagram:



The following notation for a transfer matrix has already been adopted in the previous chapters:

$$\left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right] := \mathcal{F}_u\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}, \Delta\right).$$

It is easy to see that this notation can be adopted for general dynamical systems, e.g., multidimensional systems, as far as the structure Δ is specified. This notation means that the transfer matrix can be expressed as an LFT of Δ with the coefficient matrix $\begin{bmatrix} A & B \\ C & D \end{bmatrix}$. In this special case, we say the parameter matrix Δ is the *frequency structure* of the system state space realization. This notation is deliberately somewhat ambiguous and can be viewed as both a transfer matrix and one of its realizations. The ambiguity is benign and convenient and can always be resolved from the context.

Frequency Transformation

The bilinear transformation between the z -domain and s -domain

$$s = \frac{z + 1}{z - 1}$$

transforms the unit disk to the left-half plane and is the simplest example of an LFT. We may rewrite it in our standard form as

$$\frac{1}{s}I = I - \sqrt{2}I \quad z^{-1}I \quad (I + z^{-1}I)^{-1} \quad \sqrt{2}I = \mathcal{F}_u(N, z^{-1}I)$$

where

$$N = \begin{bmatrix} I & \sqrt{2}I \\ -\sqrt{2}I & -I \end{bmatrix}.$$

Now consider a continuous system

$$G(s) = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right] = \mathcal{F}_u(M, \frac{1}{s}I)$$

where

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix};$$

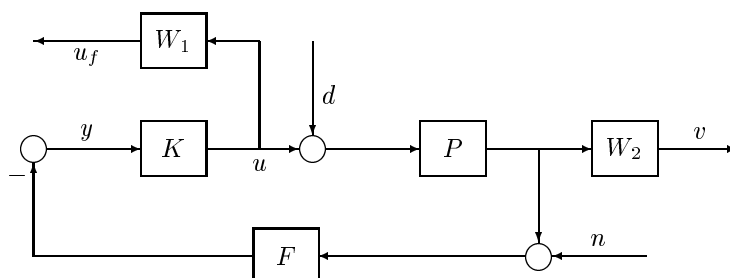
then the corresponding discrete time system realization is given by

$$\tilde{G}(z) = \mathcal{F}_u(M, \frac{z-1}{z+1}I) = \mathcal{F}_u(M, \mathcal{F}_u(N, z^{-1}I)) = \mathcal{F}_u(\tilde{M}, z^{-1}I)$$

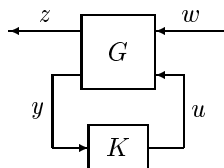
with

$$\tilde{M} = \begin{bmatrix} -(I - A)^{-1}(I + A) & -\sqrt{2}(I - A)^{-1}B \\ \sqrt{2}C(I - A)^{-1} & C(I - A)^{-1}B + D \end{bmatrix}.$$

The transformation from the z -domain to the s -domain can be obtained similarly.



A feedback system with the following block diagram can be rearranged as an LFT:


$$w = \begin{pmatrix} d \\ n \end{pmatrix} \qquad z = \begin{pmatrix} v \\ u_f \end{pmatrix}$$
$$G = \left[\begin{array}{cc|c} W_2 P & 0 & W_2 P \\ 0 & 0 & W_1 \\ \hline -F P & -F & -F P \end{array} \right].$$

Using the properties of LFTs, we can show that constrained structure control synthesis problems can be converted to constrained structure *constant* output feedback problems. Consider the synthesis structure in the last example and assume

$$G = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right] \quad K = \left[\begin{array}{c|c} A_K & B_K \\ \hline C_K & D_K \end{array} \right].$$

$$\mathcal{F}_\ell(G, K) = \mathcal{F}_\ell(M(s), F)$$

where

$$M(s) = \left[\begin{array}{cc|cc|cc} A & 0 & B_1 & 0 & B_2 & \\ 0 & 0 & 0 & I & 0 & \\ \hline C_1 & 0 & D_{11} & 0 & D_{12} & \\ 0 & I & 0 & 0 & 0 & \\ \hline C_2 & 0 & D_{21} & 0 & D_{22} & \end{array} \right]$$

and

$$F = \begin{bmatrix} A_K & B_K \\ C_K & D_K \end{bmatrix}.$$

Note that F is a *constant matrix*, not a system matrix. Hence if the controller structure is fixed (or constrained), then the corresponding control problem becomes a constant (constrained) output feedback problem.

Parametric Uncertainty: A Mass/Spring/Damper System

One natural type of uncertainty is unknown coefficients in a state space model. To motivate this type of uncertainty description, we will begin with a familiar mass/spring/damper system, shown below in Figure 10.1.

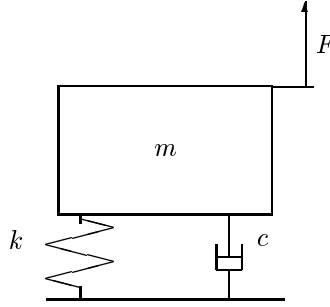


Figure 10.1: Mass/Spring/Damper System

The dynamical equation of the system motion can be described by

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = \frac{F}{m}.$$

Suppose that the 3 physical parameters m , c , and k are not known exactly, but are believed to lie in known intervals. In particular, the actual mass m is within 10% of a nominal mass, $\bar{m}$, the actual damping value c is within 20% of a nominal value of $\bar{c}$, and the spring stiffness is within 30% of its nominal value of $\bar{k}$. Now introducing perturbations δ_m , δ_c , and δ_k , which are assumed to be unknown but lie in the interval $[-1, 1]$, the block diagram for the dynamical system can be shown in Figure 10.2.

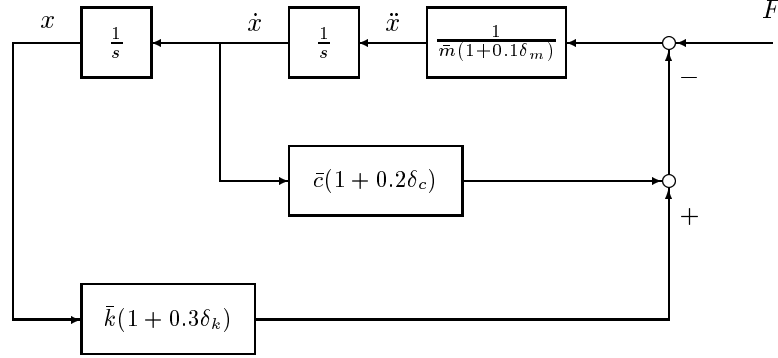


Figure 10.2: Block Diagram of Mass/Spring/Damper Equation

It is easy to check that $\frac{1}{m}$ can be represented as an LFT in δ_m :

$$\frac{1}{m} = \frac{1}{\bar{m}(1+0.1\delta_m)} = \frac{1}{\bar{m}} - \frac{0.1}{\bar{m}}\delta_m(1+0.1\delta_m)^{-1} = \mathcal{F}_\ell(M_1, \delta_m).$$

with $M_1 = \begin{bmatrix} \frac{1}{\bar{m}} & -\frac{0.1}{\bar{m}} \\ 1 & -0.1 \end{bmatrix}$. Suppose that the input signals of the dynamical system are selected as $x_1 = x, x_2 = \dot{x}, F$, and the output signals are selected as $\dot{x}_1$ and $\dot{x}_2$. To represent the system model as an LFT of the natural uncertainty parameters δ_m, δ_c and δ_k , we shall first isolate the uncertainty parameters and denote the inputs and outputs of δ_k, δ_c and δ_m as y_k, y_c, y_m and u_k, u_c, u_m , respectively, as shown in Figure 10.3. Then

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ y_k \\ y_c \\ y_m \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{\bar{k}}{\bar{m}} & -\frac{\bar{c}}{\bar{m}} & \frac{1}{\bar{m}} & -\frac{1}{\bar{m}} & -\frac{1}{\bar{m}} & -\frac{0.1}{\bar{m}} \\ 0.3\bar{k} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.2\bar{c} & 0 & 0 & 0 & 0 \\ -\bar{k} & -\bar{c} & 1 & -1 & -1 & -0.1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ F \\ u_k \\ u_c \\ u_m \end{bmatrix}, \quad \begin{bmatrix} u_k \\ u_c \\ u_m \end{bmatrix} = \Delta \begin{bmatrix} y_k \\ y_c \\ y_m \end{bmatrix}$$

i.e.,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \mathcal{F}_\ell(M, \Delta) \begin{bmatrix} x_1 \\ x_2 \\ F \end{bmatrix}$$

where

$$M = \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{\bar{k}}{\bar{m}} & -\frac{\bar{c}}{\bar{m}} & \frac{1}{\bar{m}} & -\frac{1}{\bar{m}} & -\frac{1}{\bar{m}} & -\frac{0.1}{\bar{m}} \\ \hline 0.3\bar{k} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.2\bar{c} & 0 & 0 & 0 & 0 \\ -\bar{k} & -\bar{c} & 1 & -1 & -1 & -0.1 \end{array} \right], \quad \Delta = \begin{bmatrix} \delta_k & 0 & 0 \\ 0 & \delta_c & 0 \\ 0 & 0 & \delta_m \end{bmatrix}.$$

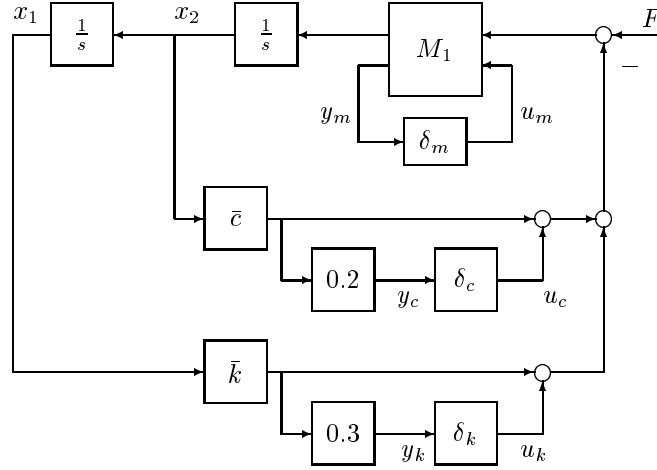


Figure 10.3: Mass/Spring/Damper System

General Affine State-Space Uncertainty

We will consider a special class of state space models with unknown coefficients and show how this type of uncertainty can be represented via the LFT formulae with respect to an uncertain parameter matrix so that the perturbations enter the system in a feedback form. This type of modeling will form the basic building blocks for components with *parametric* uncertainty.

Consider a linear system $G_\delta(s)$ that is parameterized by k uncertain parameters, $\delta_1, \dots, \delta_k$, and has the realization

$$G_\delta(s) = \left[\begin{array}{c|c} A + \sum_{i=1}^k \delta_i \hat{A}_i & B + \sum_{i=1}^k \delta_i \hat{B}_i \\ \hline C + \sum_{i=1}^k \delta_i \hat{C}_i & D + \sum_{i=1}^k \delta_i \hat{D}_i \end{array} \right].$$

Here $A, \hat{A}_i \in \mathbb{R}^{n \times n}$, $B, \hat{B}_i \in \mathbb{R}^{n \times n_u}$, $C, \hat{C}_i \in \mathbb{R}^{n_y \times n}$, and $D, \hat{D}_i \in \mathbb{R}^{n_y \times n_u}$.

The various terms in these state equations are interpreted as follows: the nominal system description $G(s)$, given by known matrices A, B, C , and D , is (A, B, C, D) and the parametric uncertainty in the nominal system is reflected by the k scalar uncertain parameters $\delta_1, \dots, \delta_k$, and we can specify them, say by $\delta_i \in [-1, 1]$. The structural knowledge about the uncertainty is contained in the matrices $\hat{A}_i, \hat{B}_i, \hat{C}_i$, and $\hat{D}_i$. They reflect how the i 'th uncertainty, δ_i , affects the state space model.

Now, we consider the problem of describing the perturbed system via the LFT formulae so that all the uncertainty can be represented as a nominal system with the unknown parameters entering it as the feedback gains. This is shown in Figure 10.4.

Since $G_\delta(s) = \mathcal{F}_u(M_\delta, \frac{1}{s}I)$ where

$$M_\delta \triangleq \begin{bmatrix} A + \sum_{i=1}^k \delta_i \hat{A}_i & B + \sum_{i=1}^k \delta_i \hat{B}_i \\ C + \sum_{i=1}^k \delta_i \hat{C}_i & D + \sum_{i=1}^k \delta_i \hat{D}_i \end{bmatrix},$$

we need to find an LFT representation for the matrix M_δ with respect to

$$\Delta_p = \text{diag}\{\delta_1 I, \delta_2 I, \dots, \delta_k I\}.$$

To achieved this with the smallest possibly size of repeated blocks, let q_i denote the rank of the matrix

$$P_i \triangleq \begin{bmatrix} \hat{A}_i & \hat{B}_i \\ \hat{C}_i & \hat{D}_i \end{bmatrix} \in \mathbb{R}^{(n+n_y) \times (n+n_u)}$$

for each i . Then P_i can be written as

$$P_i = \begin{bmatrix} L_i \\ W_i \end{bmatrix} \begin{bmatrix} R_i \\ Z_i \end{bmatrix}^*$$

where $L_i \in \mathbb{R}^{n \times q_i}$, $W_i \in \mathbb{R}^{n_y \times q_i}$, $R_i \in \mathbb{R}^{n \times q_i}$ and $Z_i \in \mathbb{R}^{n_u \times q_i}$. Hence, we have

$$\delta_i P_i = \begin{bmatrix} L_i \\ W_i \end{bmatrix} [\delta_i I_{q_i}] \begin{bmatrix} R_i \\ Z_i \end{bmatrix}^*,$$

and M_δ can be written as

$$M_\delta = \overbrace{\begin{bmatrix} A & B \\ C & D \end{bmatrix}}^{M_{11}} + \overbrace{\begin{bmatrix} L_1 & \cdots & L_k \\ W_1 & \cdots & W_k \end{bmatrix}}^{M_{12}} \overbrace{\begin{bmatrix} \delta_1 I_{q_1} & & \\ & \ddots & \\ & & \delta_k I_{q_k} \end{bmatrix}}^{\Delta_p} \overbrace{\begin{bmatrix} R_1^* & Z_1^* \\ \vdots & \vdots \\ R_k^* & Z_k^* \end{bmatrix}}^{M_{21}}$$

i.e.

$$M_\delta = \mathcal{F}_\ell \left(\begin{bmatrix} M_{11} & M_{12} \\ M_{21} & 0 \end{bmatrix}, \Delta_p \right).$$

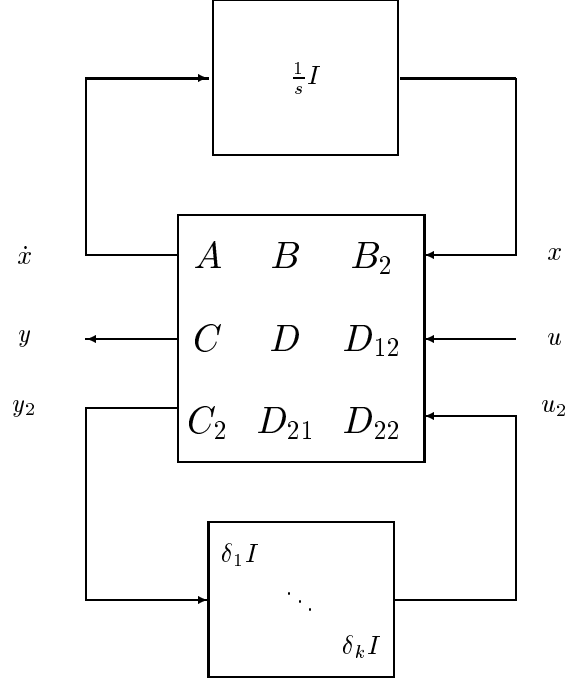


Figure 10.4: LFT Representation of State Space Uncertainty

Therefore, the matrices B_2, C_2, D_{12}, D_{21} , and D_{22} in the diagram are

$$\begin{aligned} B_2 &= \begin{bmatrix} L_1 & L_2 & \cdots & L_k \end{bmatrix} \\ D_{12} &= \begin{bmatrix} W_1 & W_2 & \cdots & W_k \end{bmatrix} \\ C_2 &= \begin{bmatrix} R_1 & R_2 & \cdots & R_k \end{bmatrix}^* \\ D_{21} &= \begin{bmatrix} Z_1 & Z_2 & \cdots & Z_k \end{bmatrix}^* \\ D_{22} &= 0 \end{aligned}$$

and

$$G_\delta(\Delta) = \mathcal{F}_u \left(\mathcal{F}_\ell \left(\begin{bmatrix} M_{11} & M_{12} \\ M_{21} & 0 \end{bmatrix}, \Delta_p \right), \frac{1}{s} I \right).$$

10.3 Basic Principle

We have studied several simple examples of the use of LFTs and, in particular, their role in modeling uncertainty. The basic principle at work here in writing a matrix LFT is often referred to as “*pulling out the Δ s*”. We will try to illustrate this with another picture (inspired by Boyd and Barratt [1991]). Consider a structure with four substructures interconnected in some known way as shown in Figure 10.5.

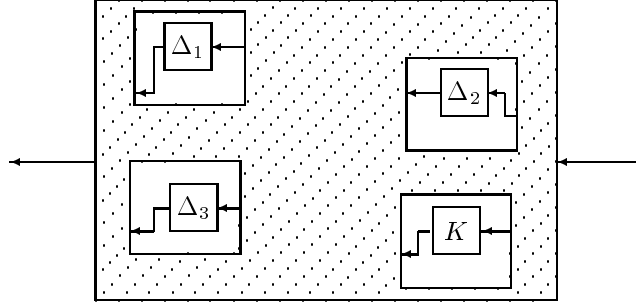


Figure 10.5: Multiple Source of Uncertain Structure

This diagram can be redrawn as a standard one via “pulling out the Δ s” in Figure 10.6. Now the matrix “ M ” of the LFT can be obtained by computing the corresponding transfer matrix in the shadowed box.

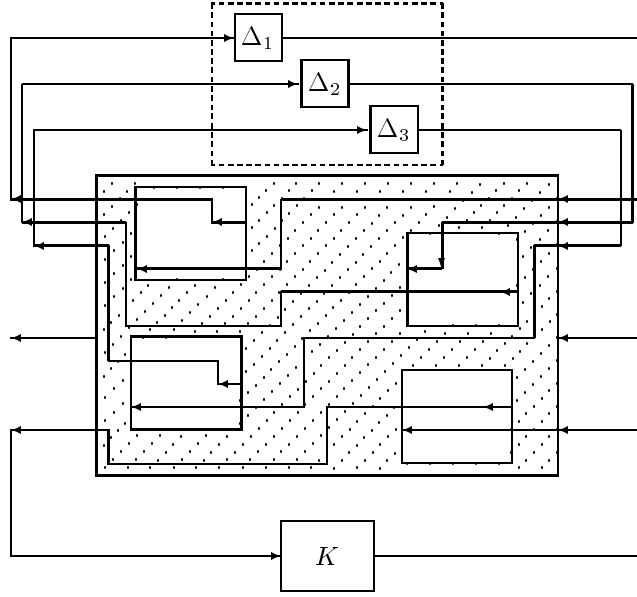
We shall illustrate the above principle with an example. Consider an input/output relation

$$z = \frac{a + b\delta_2 + c\delta_1\delta_2^2}{1 + d\delta_1\delta_2 + e\delta_1^2}w =: Gw$$

where a, b, c, d and e are given constants or transfer functions. We would like to write G as an LFT in terms of δ_1 and δ_2 . We shall do this in three steps:

1. Draw a block diagram for the input/output relation with each δ separated as shown in Figure 10.7.
2. Mark the inputs and outputs of the δ 's as y 's and u 's, respectively. (This is essentially *pulling out the δ s*).
3. Write z and y 's in terms of w and u 's with all δ 's taken out. (This step is equivalent to computing the transformation in the shadowed box in Figure 10.6.)

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ z \end{bmatrix} = M \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ w \end{bmatrix}$$

Figure 10.6: Pulling out the Δ s

where

$$M = \left[\begin{array}{ccc|c|c} 0 & e & d & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & be & bd + c & 0 & b \\ \hline 0 & ae & ad & 1 & a \end{array} \right].$$

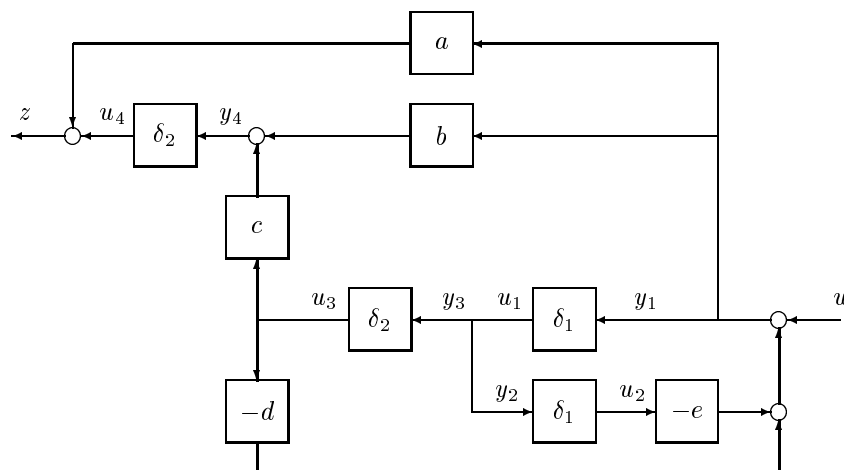
Then

$$z = \mathcal{F}_u(M, \Delta)w, \quad \Delta = \begin{bmatrix} \delta_1 I_2 & 0 \\ 0 & \delta_2 I_2 \end{bmatrix}.$$

All LFT examples in the last section can be obtained following the above steps.

10.4 Redheffer Star-Products

The most important property of LFTs is that any interconnection of LFTs is again an LFT. This property is by far the most often used and is the heart of LFT machinery. Indeed, it is not hard to see that most of the interconnection structures discussed early, e.g., feedback and cascade, can be viewed as special cases of the so-called *star product*.



Suppose that P and K are compatibly partitioned matrices

such that the matrix product $P_{22}K_{11}$ is well defined and square, and assume further that $I - P_{22}K_{11}$ is invertible. Then the *star product of P and K with respect to this partition* is defined as

Note that this definition is dependent on the partitioning of the matrices P and K above. In fact, this star product may be well defined for one partition and not well defined for another; however, we will not explicitly show this dependence because it is always clear from the context. In a block diagram, this dependence appears, as shown in Figure 10.8.

$$P = \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right] \quad K = \left[\begin{array}{c|cc} A_K & B_{K1} & B_{K2} \\ \hline C_{K1} & D_{K11} & D_{K12} \\ C_{K2} & D_{K21} & D_{K22} \end{array} \right].$$
$$\mathcal{S}(P, K): \begin{bmatrix} w \\ \hat{w} \end{bmatrix} \mapsto \begin{bmatrix} z \\ \hat{z} \end{bmatrix}$$

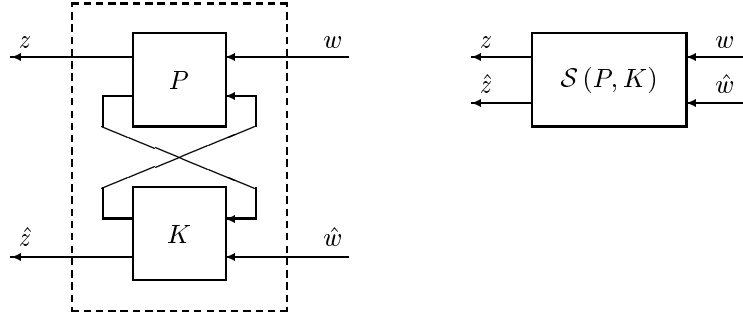


Figure 10.8: Interconnection of LFTs

has a representation

$$S(P, K) = \left[\begin{array}{c|cc} \bar{A} & \bar{B}_1 & \bar{B}_2 \\ \hline \bar{C}_1 & \bar{D}_{11} & \bar{D}_{12} \\ \bar{C}_2 & \bar{D}_{21} & \bar{D}_{22} \end{array} \right] = \left[\begin{array}{c|c} \bar{A} & \bar{B} \\ \hline \bar{C} & \bar{D} \end{array} \right]$$

where

$$\begin{aligned} \bar{A} &= \begin{bmatrix} A + B_2 \tilde{R}^{-1} D_{K11} C_2 & B_2 \tilde{R}^{-1} C_{K1} \\ B_{K1} R^{-1} C_2 & A_K + B_{K1} R^{-1} D_{22} C_{K1} \end{bmatrix} \\ \bar{B} &= \begin{bmatrix} B_1 + B_2 \tilde{R}^{-1} D_{K11} D_{21} & B_2 \tilde{R}^{-1} D_{K12} \\ B_{K1} R^{-1} D_{21} & B_{K2} + B_{K1} R^{-1} D_{22} D_{K12} \end{bmatrix} \\ \bar{C} &= \begin{bmatrix} C_1 + D_{12} D_{K11} R^{-1} C_2 & D_{12} \tilde{R}^{-1} C_{K1} \\ D_{K21} R^{-1} C_2 & C_{K2} + D_{K21} R^{-1} D_{22} C_{K1} \end{bmatrix} \\ \bar{D} &= \begin{bmatrix} D_{11} + D_{12} D_{K11} R^{-1} D_{21} & D_{12} \tilde{R}^{-1} D_{K12} \\ D_{K21} R^{-1} D_{21} & D_{K22} + D_{K21} R^{-1} D_{22} D_{K12} \end{bmatrix} \\ R &= I - D_{22} D_{K11}, \quad \tilde{R} = I - D_{K11} D_{22}. \end{aligned}$$

In fact, it is easy to show that

$$\begin{aligned} \bar{A} &= S \left(\begin{bmatrix} A & B_2 \\ C_2 & D_{22} \end{bmatrix}, \begin{bmatrix} D_{K11} & C_{K1} \\ B_{K1} & A_K \end{bmatrix} \right), \\ \bar{B} &= S \left(\begin{bmatrix} B_1 & B_2 \\ D_{21} & D_{22} \end{bmatrix}, \begin{bmatrix} D_{K11} & D_{K12} \\ B_{K1} & B_{K2} \end{bmatrix} \right), \end{aligned}$$

$$\begin{aligned}\bar{C} &= S\left(\begin{bmatrix} C_1 & D_{12} \\ C_2 & D_{22} \end{bmatrix}, \begin{bmatrix} D_{K11} & C_{K1} \\ D_{K21} & C_{K2} \end{bmatrix}\right), \\ \bar{D} &= S\left(\begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix}, \begin{bmatrix} D_{K11} & D_{K12} \\ D_{K21} & D_{K22} \end{bmatrix}\right).\end{aligned}$$

10.5 Notes and References

This chapter is based on the lecture notes by Packard [1991] and the paper by Doyle, Packard, and Zhou [1991].